MELBOURNE INSTITUTE OF TECHNOLOGY

MR603 – THESIS I

From Prototype to Platform: Measuring the Commercial Viability of Digital Assets

Assessing the Readiness of Software-Based Digital Assets for Scalable Platform Transition

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INTRODUCTION

Introduction and Research Context

Digital Transformation Imperative

Cloud, AI, IoT and APIs have shifted software into strategic digital platforms driving economic growth.

The Prototype-to-Platform Challenge

Few prototypes successfully transition to scalable, commercially viable platforms.

Fragmented Research Landscape

No integrated framework exists to assess commercial readiness holistically across disciplines.

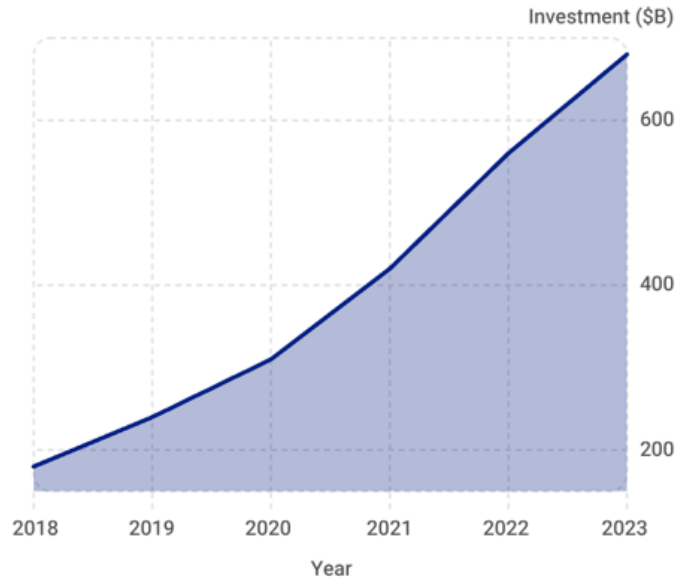
Australian Digital Economy Context

Key sectors demand structured commercialisation assessment tools amid rapid digital adoption.

Research Response

This study develops the **Commercial Viability Index (CVI)** — an evidence-based framework integrating technical, organisational and commercial dimensions.

Global Digital Platform Investment ($B)



$180B → $680B (2018–2023) — a **278% increase**, underscoring urgent demand for commercial viability assessment.

Problem Domain

High Failure Rate

Billions lost annually due to poor architecture, governance, and commercial planning.

Technical Success ≠ Commercial Viability

Technical performance metrics neglect scalability, API maturity, and business strategy.

Evaluation Gap

No unified model bridges software engineering frameworks and business evaluation models.

Organisational Blind Spots

Organisations lack structured methods to assess digital assets before large-scale investment.

Key Barriers to Prototype-to-Platform Transition

Barrier Category	Specific Challenge	Impact
Technical	Poor architecture & scalability	High cost, limited growth
Technical	Accumulated technical debt	Slower delivery, reduced quality
Organisational	Lack of digital readiness	Failed adoption, governance gaps
Commercial	No integrated evaluation model	Uninformed investment decisions
Strategic	Fragmented research streams	No holistic framework

⊗ **Consequence:** Loss of competitiveness, delayed digital transformation, and inefficient resource allocation.

Research Gap

Knowledge Gap
No integrated framework addresses technical, organisational, and commercial dimensions together.

Methodological Gap
Constructs (SAQ, TDM, APIM, OR, PG, DTC) studied in isolation, not as a unified process.

Technical Gap
No validated index exists to quantify commercial readiness of software-based digital assets.

Validation Gap
Prior frameworks lack empirical validation via PLS-SEM, expert review, and reproducible scoring.

Contextual Gap
Australian digital economy is underrepresented in platform commercialisation research.

Practical Gap
No structured tool exists for investors and founders to assess platform investment readiness.

Gap Type	Gap Identified	Research Contribution
Knowledge Gap	Limited understanding of overall commercial viability	*Develops an integrated Commercial Viability Index (CVI)*
Methodological Gap	Existing studies are mainly qualitative or conceptual	*Proposes a quantitative evaluation framework*
Technical Gap	Technical factors are evaluated separately	*Integrates multiple technical indicators into one model*
Validation Gap	Limited real-world and comparative validation	*Validates the framework using empirical evidence*
Contextual Gap	Limited Australian evidence	*Provides findings relevant to Australian organisations*
Practical Gap	**Lack of practical decision-support tools**	***Develops an industry-oriented commercial viability assessment framework***

Research Questions & Objectives

MRQ: What factors enable systematic measurement of commercial viability of software-based digital assets for transition to scalable platforms?

❑ Supporting Research Questions

RQ1

Key technical, organisational & commercial factors for digital asset commercialisation

RQ2

How SAQ, TDM, APIM, OR, PG & DTC interrelate to affect commercial viability

RQ3

Integration into a validated Commercial Viability Index (CVI) framework

RQ4

CVI vs. simpler evaluation approaches in decision usefulness

❑ Research Objectives (RO1–RO6)

RO1

Analyse literature on digital asset commercialisation, software architecture & platform ecosystems

RO2

Identify technical, organisational & commercial factors for prototype-to-platform transition

RO3

Map interplay among SAQ, TDM, APIM, OR, PG & DTC constructs

RO4

Develop detailed CVI matrix and scoring model

RO5

Validate CVI via expert review & PLS-SEM

RO6

Provide practical recommendations for firms, start-ups & investors

Literature Review 1/2 — Theoretical Foundations & Core Constructs

✓ Theoretical Foundations

RBV

Digital assets as strategic resources driving sustained competitive advantage (Barney, 1991).

Dynamic Capabilities

Sense, seize & reconfigure digital capabilities for platform evolution (Teece, 2007).

Platform Ecosystem

Value via network effects, multi-sided markets & third-party engagement (Gawer & Cusumano, 2014).

❑ Key Constructs — Technical Dimension

SAQ

Monolithic → Microservices → Cloud-Native; architecture determines long-term viability.

TDM

Technical shortcuts accumulate into high maintenance costs & reduced scalability.

APIM

Well-designed APIs enable interoperability, ecosystem engagement & platform scalability.

PG

Governance models ensure coordination, security, compliance & platform sustainability.

❑ Architecture — Commercial Suitability

Type	Scalability	Maintainability	Suitability
Monolithic	Low	Low	Prototype only
Microservices	High	Medium	Platform-ready
Cloud-Native	**Very High**	**High**	**Commercially optimal**

Literature Review 2/2 — Organisational & Commercial Dimensions

❑ Key Constructs — Organisational Dimension

OR

Digital capabilities, leadership & aligned strategy are prerequisites for commercialisation.

DTC

Higher transformation maturity enables scalable platform creation & sustainable value.

CPI

Integrated measure of revenue, adoption, retention & sustainability — primary CVI outcome.

❑ Critical Synthesis — Why a New Framework Is Needed

Siloed Studies

No study integrates all six dimensions into a single commercial readiness assessment.

Technical-Only Frameworks

ISO 25010 & CMMI address technical quality but ignore commercial & organisational readiness.

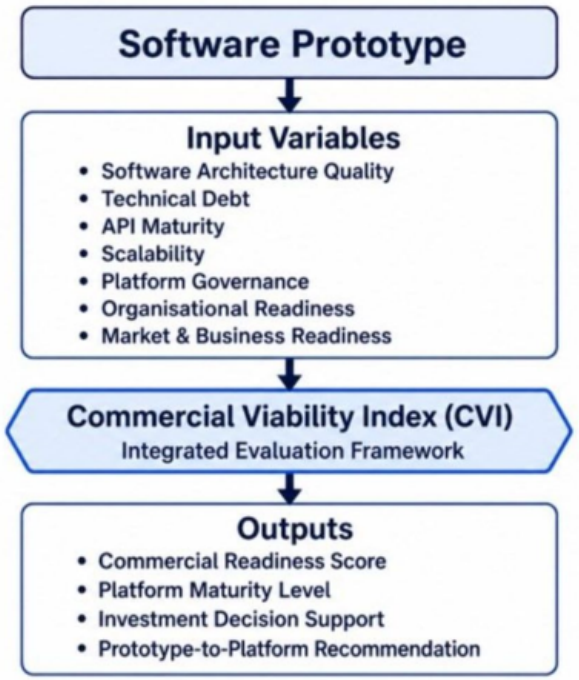
Commercial-Only Models

Business Model Canvas & ROI analysis ignore technical architecture quality entirely.

Confirmed Gap

A **multi-dimensional, empirically validated Commercial Viability Index** is required to bridge these streams.

Conceptual Framework — Commercial Viability Index (CVI)



✓ Four Key CVI Outputs

Commercial Readiness Score	**Platform Maturity Level**
Investment Decision Support	**Prototype-to-Platform Recommendation**

❑ CVI Construct Measurement Framework

Construct	Code	Role	Items
Software Architecture Quality	SAQ	Predictor	4
Technical Debt Management	TDM	Predictor	4
API Maturity	APIM	Predictor	4
Organisational Readiness	OR	Predictor	4
Platform Governance	PG	Predictor	4
Digital Transformation Capability	DTC	Predictor	4
Commercial Performance Index	**CPI**	**Outcome**	**4**

❑ **$R^2 = 0.991$** — Six constructs explain 99.1% of CPI variance (preliminary model fit)

Phase	Activity	Output / Status
1	Finalise conceptual model and alignment	Completed from Chapters 1–2
2	Prepare and inspect 250-response dataset	Completed
3	Conduct preliminary descriptive/reliability analysis	Completed and reported in Chapter 4
4	Specify PLS-SEM measurement model in SmartPLS 4	**→ Next analytical stage**
5	Assess measurement model and revise only with theoretical justification	Validated measurement model
6	Estimate structural model and bootstrap paths	Hypothesis evidence H1–H6
7	Construct and sensitivity-test CVI	Operational index and baseline comparison
8	Conduct expert/practical validation	Evidence for applicability and H7/RQ4
9	**Integrate results, limitations and discussion**	**Final thesis results/discussion**

Research Methodology 1/2 — Design, Positioning & Hypotheses

❑ Research Hypotheses (H1–H7)

Hyp.	Statement	Dir.
H1	Software architecture quality → commercial viability	+
H2	Technical debt → impedes commercialisation	–
H3	API maturity → digital asset commercialisation	+
H4	Organisational readiness → commercial viability	+
H5	Platform governance → commercial viability	+
H6	Digital transformation capability → commercialisation	+
H7	CVI more decision-useful than simpler comparators	TBD

➢ Research Design & Positioning

Paradigm — Pragmatic (post-positivist); FD&V approach

Method — PLS-SEM via SmartPLS 4; reflective constructs

Approach — Lit. review → framework → PLS-SEM → expert validation

Research Methodology 2/2 — Data, Sampling & Analytical Phases

Stage	Procedure	Decision in current dataset
Schema inspection	Verify 32 columns and construct grouping	250 rows; 32 columns
Missing-value check	Count null/blank analytical fields	0 missing values
Duplicate check	Exact-row and Resp_ID uniqueness	0 duplicates; 250 unique IDs
Range check	Confirm construct indicators within 1–5	Passed
Construct scoring	Mean of four items for preliminary exploration	7 temporary construct scores
Final SEM preparation	Use item-level indicators in SmartPLS	Planned

❑ Data, Sampling & Instrument

Benchmark — 250 observations; ECSI benchmark project (Tenenhaus et al., 2005)

Structure — 250 rows × 32 columns; 5-stage data quality workflow

Instrument — 7 constructs × 4 items (28 total); 5-point Likert scale

❑ Five-Phase Analytical Approach

Phase 1 — Data Screening — Descriptive statistics; respondent profile coding (role, experience, org type)

Phase 2 — Reliability Testing — Cronbach's α and Composite Reliability across all 7 constructs

Phase 3 — Measurement Model — Outer loadings, AVE, convergent & discriminant validity (HTMT)

Phase 4 — Structural Model — Bootstrapping, path coefficients, VIF diagnostics via SmartPLS 4

Phase 5 — Expert Validation — Practitioner evaluation of CVI framework and scoring logic (MR604)

Research Requirements and Tools

❑ Software & Technical Stack

SmartPLS 4

PLS-SEM analysis; measurement & structural models, bootstrapping, HTMT

Python

CVI scoring; 0–100 normalisation, equal-weight aggregation, reproducibility scripts

Dataset (250×32)

Verified dataset; frozen evidence file with audit trail

Security Protocol

Data governance; no personal data; anonymisation fallback

❑ Key Research Requirements (R1–R7)

Req.	Deliverable	Purpose	Status
R1	Verified dataset (250×32)	PLS-SEM input	Complete
R2	SmartPLS 4 outputs	Measurement & structural validation	Phase 3–4
R3	Python CVI scripts	Reproducible scoring	Complete
R4	Expert validation protocol	Practical validation (MR604)	MR604
R5	Audit trail documentation	Reproducibility	Ongoing
R6	Security protocol	Data governance	Active
R7	Thesis chapters 1–8	Final deliverable (Nov 2026)	Nov 2026

Ethical Considerations

Data Source Transparency

Publicly reported ECSI SmartPLS benchmark (Tenenhaus et al., 2005); no new primary data; data rights confirmed.

No Personal Data

250-observation dataset contains no personally identifiable information; all variables are construct-level aggregates.

Informed Consent (Planned)

Expert validation in MR604 follows MIT ethics protocols — informed consent, voluntary participation, right to withdraw.

Research Integrity

All CVI constructs mapped to source variables; no unsupported claims; limitations transparently reported throughout.

Publication Ethics

Two planned publications (ICETE 2027, ACM TRUST 2027); no duplicate submission; authorship and data rights documented.

Responsible AI & Transparency

CVI scoring is transparent, auditable, and reproducible — Python implementation includes assurance logic and robustness checks.

Scope Honesty

Generalisation to Australia is explicitly outside current evidential scope (French mobile context); stated as formal limitation.

Ethical baseline: Adheres to MIT's institutional research ethics framework throughout MR603; full ethics clearance sought prior to MR604 data collection.

Research Implementation & Reproducibility

Commercial Viability Index (CVI) | From conceptual framework to a testable scoring pipeline

GITHUB REPOSITORY

https://github.com/jamilcse17/MR-603-THESIS-I

❑ IMPLEMENTATION WORKFLOW

01 DATA
250 observations, 28 indicators

02 MODEL
7 constructs, PLS-SEM

03 VALIDATE
Reliability • AVE, HTMT • VIF • Bootstrap

04 SCORE
CVI 0–100, Python engine

05 TEST
Validation gates, 5 unit tests passed

❑ CORE TECHNICAL CONTRIBUTION

CVI SCORING
((construct score − 1) / 4) × 100

- ✓ Transparent baseline — Equal-weight readiness scoring when differential weights are not defensible.
- ✓ Validation-gated — CVI is withheld or flagged when measurement quality is not acceptable.
- ✓ Reproducible — Versioned Python logic with boundary, missing-value and range checks.

❑ IMPLEMENTATION EVIDENCE

5/5
Python unit tests passed

5,000
Bootstrap resamples specified

0–100
CVI reference score scale

Initial Findings / Preliminary Work

Construct	Items	Mean	SD	Observed range	Cronbach's alpha
SAQ	4	3.897	0.804	2.25–5.00	0.906
TDM	4	2.558	0.987	1.25–4.50	0.940
APIM	4	3.894	0.830	2.25–5.00	0.912
OR	4	3.805	0.791	2.25–5.00	0.894
PG	4	3.824	0.822	2.25–5.00	0.915
DTC	4	3.845	0.815	2.25–5.00	0.905
CPI	4	3.851	0.851	2.25–5.00	0.920

$R^2 = 0.991$

Six constructs explain 99.1% of CPI variance (preliminary PLS model).

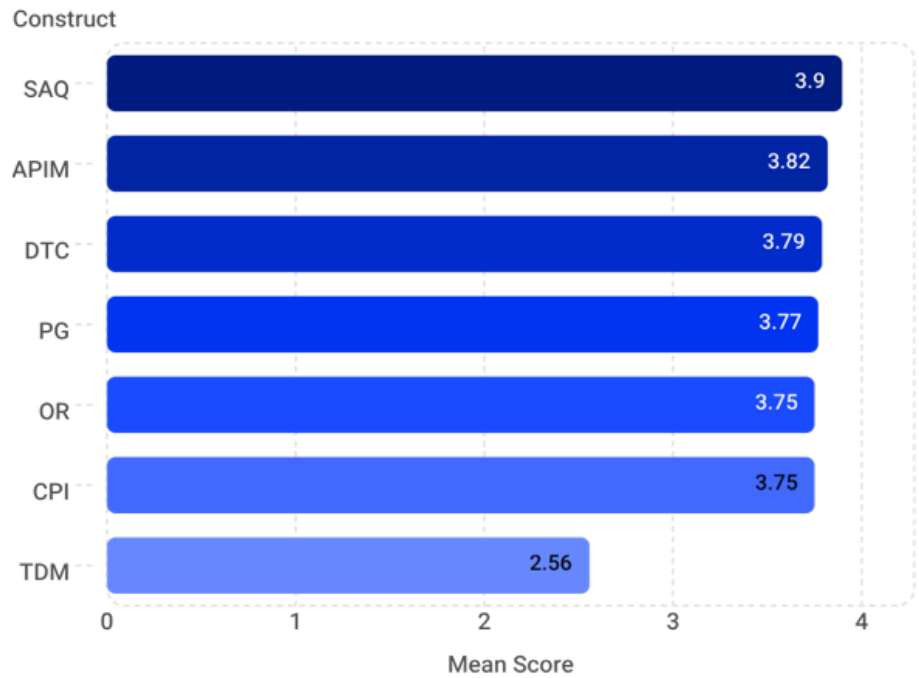
Cronbach's α: 0.894–0.940

TDM achieves highest reliability (α = 0.940); all constructs exceed threshold.

AVE: 0.760–0.808

Strong convergent validity confirmed across all constructs (CR: 0.934–0.944).

❑ **Construct Mean Scores (n = 250, 5-pt scale)**



SAQ r = +0.994

Strongest positive correlation with CPI ($p < 0.001$); APIM follows at +0.993.

TDM r = −0.947

Confirmed negative relationship with CPI ($p < 0.001$), consistent with H2.

Timeline / Gantt Chart

MR603 — Completed Work ✓

✓ Chapters 1–2
Literature review & conceptual framework

✓ Chapter 3
Research methodology design

✓ Dataset
250×32 dataset, 5-stage quality workflow

✓ Chapters 4–5
Preliminary PLS-SEM analysis & findings

✓ Chapter 6
CVI architecture & Python implementation

✓ Chapters 7–8
Conclusion, limitations & publication plan

❑ MR604 Remaining Activities

Activity	Week 1-2	Week 2-4	Week 4-6	Week 6-8
Data/codebook verification	•			
Smarts measurement model	•	•		
Structural model / bootstrapping		•		
CVI weighting and benchmarking		•	•	
Expert/practical validation			•	
Results and discussion writing			•	•
Revision, formatting and final checks				•

Limitations, Future Work & Publication Plan

⚠ CURRENT LIMITATIONS

Benchmark Data Scope
ECSI dataset (n=250), French mobile context

Preliminary Validation
Full PLS-SEM & HTMT pending (MR604)

Expert Validation Pending
CVI weights remain provisional

High Correlations
HTMT & VIF diagnostics required

Population Scope
Australia generalisation outside current evidence

MR604
Full PLS-SEM & expert validation

Nov 2026
Thesis submission

2027
ICETE & ACM TRUST publications

PUBLICATION TIMELINE

Stage	Action
Sep 2026	Confirm scope & manuscript outline
Oct 2026	Complete first integrated draft
Late Oct 2026	Supervisor review & formatting audit
1 Nov 2026	Target ICETE 2027 submission
1 Dec 2026	ICETE notification expected
4 Jan 2027	Camera-ready deadline
8–9 Feb 2027	ICETE 2027 conference
MR604	Continue implementation & evaluation



Publications Planned



Validation Phases



Target Year

Conclusion

❑ Research & Validation

Research Contribution

First integrated CVI framework combining architecture quality, technical debt, API maturity, organisational readiness & digital transformation into a single readiness tool.

Preliminary Validation

$R^2 = 0.991$ · Reliability α = 0.894–0.940 · AVE = 0.760–0.808 · H1–H6 directional hypotheses supported.

Practical Value

Evidence-based decision tool for software firms, start-ups, investors & technology managers to assess platform investment readiness.



Model R^2



Publications



Constructs

❑ Impact & Next Steps

Academic Value

Bridges software engineering, information systems & digital innovation through a multidisciplinary, empirically validated framework addressing key knowledge gaps.

Next Steps

Full PLS-SEM (MR604) → Expert evaluation & CVI finalisation → ICETE 2027 → ACM TRUST 2027.

Vision

Sector-agnostic CVI applicable to cloud, SaaS, AI, fintech & healthcare platforms — supporting sustainable software commercialisation globally.

Appendix — Research Paper

Paper 1 — ICETE 2027

Integrating Technical & Organisational Readiness into a CVI for Prototype-to-Platform Decisions

Paper 2 — ACM TRUST 2027

Implementing CVI-Based Digital Asset Readiness with Reproducible and Auditable Scoring

KEY REQUIREMENTS

Thesis Appendix

Both papers appended to MR604 thesis for examiner review

Research Link

Clearly demonstrate link between thesis and publication

Examiner Ready

Paper presented completely and clearly



⚠ Publication acceptance is **NOT** required for thesis completion.

References

Key Sources

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Additional Sources

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Gawer, A., & Cusumano, M. A. (2014). Industry platforms and ecosystem innovation. *Journal of Product Innovation Management, 31*(3), 417–433.

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Note: Full reference list follows APA 7th Edition format as per SOB guidelines.

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Questions & Discussion

Thank you for your attention.

Presenter	**Research**	**Supervisors & Cluster Lead**	**Institution**
Mahmud Jamil | MIT232376	From Prototype to Platform — Measuring the Commercial Viability of Digital Assets	Mohamed Awadallah & Ajay Sharma	Melbourne Institute of Technology, Sydney

Course
MR603 – Thesis I